

CM2030 Graphics Programming

Module Description

This module is aimed at programmers that have learned the basics of programming using the p5.js programming frameworks through Introduction to Programming part one.

Learners will begin by looking into building their own physics engine for animating elements on the screen, using vectors and forces. Having learned the basics of such a system they will then look into an off-the-shelf physics library. After that, learners will look into the principles of procedural generation of graphics and will hear first-hand accounts from a game designer and a computational artist.

In the second part of the module attention is shifted to 3D graphics at first before looking deeper into the basic elements of images, namely pixels. Once learners have grasped the basics of pixel reading and manipulation, they will use these techniques in combination with others to programme computer vision applications and image processing filters. The second part of the module finishes with an interview with an accomplished artist-hacker who uses computer vision techniques for a lot of his work.

Learners get to practice all the above in multiple, weekly, fun, programming exercises and their knowledge is tested through engaging programming assignments and quizzes.

Course Goals and Objectives

Upon successful completion of this course, you will be able to:

1. Manipulate numerical representations of visual media
2. Explain the basic mathematical and theoretical principles of visual computing
3. Write basic but complete graphics software systems
4. Apply filters on images in order to transform them for processing in computer vision scenarios
5. Explain the fundamental tenets of computer vision and write code that processes images to make intelligent assertions about their content
6. Select and use software libraries for software development

Textbook and Readings

Specific essential readings for each week from the following list are included in the Readings page for each week. During each week, your instructor will post links of interest and things he would like you to explore.

There is no official textbook or readings as most of the module is very practical and video based.

Course Outline

The course consists of ten topics divided into 20 weeks that focus on key concepts.

Topic 1: Introduction to graphics, vectors, and animation	Key concepts: You will learn about 2D transformations, vectors, and animation. You will see how they can be used to program a basic clock, and you will use them to program a solar system. Learning outcomes: • Explain how transformations work • Describe how classes work • Use transformations to program a basic solar system • Describe how vectors work • Apply vector arithmetic • Implement simple systems that use vectors
Topic 2: Forces and physics	Key concepts: You will learn about forces, gravity, friction, particles, and collision detection. You will be introduced to the matter.js library. Learning outcomes: • Explain how forces work • Use physics concepts in animation scenarios • Implement simple physics systems • Describe what physics engines are and what they do • Describe the basic elements of matter.js • Implement simple physics systems using matter.js

Topic 3: Physics with matter.js and generative techniques	Key concepts: You will learn about physics engines, matter.js, bodies, constraints, and composites. You will also explore generative art techniques. Learning outcomes: • Describe what physics engines are and what they do • Describe the basic elements of matter.js • Implement simple physics systems using matter.js • Explain what generative art and design is. Identify important characteristics of generative art. • Apply randomness and noise to create simple generative systems
Topic 4: Generative techniques: randomness, noise, and fractals	Key concepts: You will learn about generative art, randomness, and noise. You will also learn about generative art and design, polar coordinates, oscillation, additive synthesis, fractals, and recursion. Learning outcomes: • Explain what generative art and design is • Identify important characteristics of generative art • Apply randomness and noise to create simple generative systems • Use trigonometry to make shapes • Use oscillation to code movement • Implement generative systems using additive synthesis • Implement recursive systems to make fractals

Topic 5: Generative techniques and interviews	Key concepts: You will learn about generative art and design, polar coordinates, oscillation, additive synthesis, fractals, and recursion. You will also learn about procedural generation in games, generative art, and design. Learning outcomes: • Use trigonometry to make shapes use oscillation to code movement • Implement generative systems using additive synthesis • Implement recursive systems to make fractals • Explain what procedural generation for games is and what the pros and cons are • Identify ways Andy Lomas' work is linked to the syllabus
Topic 6: 3D graphics	Key concepts: You will learn about 3D renderer, lights, material, texture, and camera. Learning outcomes: • Explain the difference between 2D and 3D graphics • Use 3D primitives, lights, and materials to program simple animations • Implement 3D animations by manipulating the camera parameters

Topic 7: Colours, images, and pixels	Key concepts: You will learn about colours, colour theory, images, webcams, and pixels. Learning outcomes: • Explain the difference between bitmap and vector graphics • Use principles of colour theory to generate colour procedurally • Use images from the hard disk or from the webcam • Access and manipulate pixels directly
Topic 8: Image processing	Key concepts: You will learn about image processing, filters, and convolution. Learning outcomes: • Explain what image processing is and what its applications are use simple to manipulate images • Use convolution to apply advanced filters on images • Implement an Instagram-type image processing filter
Topic 9: Computer vision - part 1 and 2	Key concepts: You will learn about computer vision, colour tracking, background subtraction and frame differencing. Learning outcomes: • Explain what computer vision is and what its applications and challenges • Describe what colour tracking and background subtraction are • Use frame differencing to track movement • Explore advanced techniques such as optical flow and face tracking • Use a computer vision library

Topic 10: Computer vision - part 3 and interviews	Key concepts: You will learn about OpenCV, points to track, face recognition and optical flow. Learning outcomes: • Explain why we would use a computer vision library • Describe some advanced computer vision algorithms • Use optical flow to control objects on the canvas • Use a computer vision library
--	---

Learning Activities of This Course

The courses comprised of the following elements:

- **Lecture videos.** In each week the concepts you need to know will be presented through a collection of short video lectures. You may stream these videos for playback within the browser by clicking on their titles or download the videos.
- **Practice Quizzes.** Topics include practice quizzes, intended for you to assess your understanding of the content. You will be allowed unlimited attempts at each practice quiz. There is no time limit on how long you take to complete each attempt at the quiz. These quizzes do not contribute toward your final score in the class.
- **Practice Programming exercises.** Every week you will apply what you learned by completing short, engaging programming exercises. These assignments do not contribute toward your final score in the class.
- **Graded Assignments.** There are two graded assignments, each is worth 50% of the final module grade. All assignments will be graded by the project tutors.
- **Discussion Prompt.** Topics also include discussion prompts. You will see the discussion prompt alongside other items in the lesson. Each prompt provides a space for you to respond. After responding, you can see and comment on your peers' responses. All prompts and responses are also accessible from the general discussion forum and the topic discussion forum.
- **Readings.** Topics may include several suggested readings. They are good supplementary materials for you to further understand the module topics.

How to Pass This Course

The module will contain a range of summative and formative assessments. Summative assessments are assessments which contribute directly towards your final grade. Formative assessments do not count directly towards your final grade. Instead, they provide you with opportunities for low stakes practice and will often provide some sort of feedback about your progress. For example, a practice quiz might provide you with feedback about why a particular answer was wrong.

The course has two major assessments each worth 50% of your grade:

- Coursework 1: this consists of a project which will be assessed halfway through course (approximate deadline in week 13)
- Coursework 2: this also consists of a project which will be assessed at the end of the course (approximate deadline in week 22).

This is a detailed breakdown of all the marks:

Activity	Required? (Summative)	Deadline week	Estimated time per course	% of final grade
Written, staff graded coursework 1	Yes	Approximately week 13	20 hours	50%
Written, staff graded coursework 2	Yes	Approximately week 22	20 hours	50%